Core Mathematics

A Practical Guide on Mathematics for Physics Learners

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Lecture notes for first-year university students in physics or physical science, covering mathematical tools you the students need most. Calculus, logic, complex numbers, and vectors, with introductions to vector calculus, differential equations, and statistical analysis. Exercises are central to these notes. Practice each topic repeatedly until you can work through problems quickly and accurately.

Preface

Physics is an activity to describe the world, but how? At present, mathematics (+ *broken* English) is the only language we can use. So, *unfortunately*, we need to learn mathematics! (and English...🥺) But it means we are *fortunate*: we have a language to describe the world. In fact, physics have developed together with mathematics. We understand physics better if we study mathematics well!

When you do DIY, you need to use the tools properly, smoothly, and accurately. It is the same here; when we use math as a tool, it must be **logical**, **quick**, and **accurate**. To be logical, you must think each problem carefully; otherwise your discussion might be incomplete and you can't convince others. To be quick and accurate, you must *"drill"* repeatedly, as if top NBA players do shooting practice almost everyday. You should reach the point where basic calculations feel automatic.

This is not a math textbook. **This is rather** a math "drill" book. Rigorous proofs are mostly omitted, but drill problems are the core of this document, through which students are expected to achieve better conceptual understanding.

Target of this document

This document is primarily for first-year undergraduate students in their second semester, who

- completed basic calculus (differentiation and integration),
- are going to major in physics or physical science, and
- do not hesitate to do repetitive practice to achieve better conceptual understanding.

How to use this document

1. **Buy a A4-sized paper notebook.**

   Digital notebooks on tablets are not suitable (less effective).

2. Read the text and solve **Quizzes**.

   The texts are kept short, so please read all the text carefully. Quizzes should be solved while reading, but you don't have to write down the solution process for Quizzes.

3. Solve **Drill Problems**.

   This is to improve your fluency ($\neq$ speed) and accuracy. Write down solution process clearly.

4. Solve other **Problems**.

   This is to have better conceptual understanding, but also to develop your writing skills. Write down the solution process clearly *as an English text*, taking care of *logical completeness*.

Optionally, you you are encouraged to prepare a reference book, so that you can consult it when you want to know more. Sho supposes [Boas] listed below is the best option.

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Visit https://github.com/misho104/LecturePublic for further information, updates, and to report issues.

References

This document is written under the influence of the following references.

- The Math Book by Hal Tasaki (数学：物理を学び楽しむために, in Japanese)

  A free book with full description, but unfortunately in Japanese. Topic selections and the depth of descriptions are based on this reference.

- First-Year Math Practice (in Japanese)

  This document is grounded in a "learn steadily" approach ("じっくり" in Japanese), which originates from Prof. Gocho and Prof. Kiyono, who were instructors of Sho in his freshman. Problems and rigorous math descriptions are taken from their works.

- **[Boas]** Mary L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd ed., Wiley, 2006.

- **[AWH]** George B. Arfken, Hans J. Weber, and Frank E. Harris,
  Mathematical Methods for Physicists, 7th ed., Academic Press, 2023.

  These two books are widely used in physics departments around the world. [Boas] seems more friendly to beginners, while [AWH] seems more detailed and complete. Sho recommends you to study [Boas] *in parallel with* this document, using it as a reference, and to consult [AWH] when you want more details or more exercises.

Lecture Plan

The content is designed for 150-minute × 14-week lectures, as it is originally for a lecture *Mathematics for Fundamental Physics* (基礎物理數學) in National Sun Yat-sen University (國立中山大學).

1. Derivative
2. Units. Significant Digits.
3. Logic. Theorems. a^x and $\log_a x$.
4. Logic. Theorems. a^x and $\log_a x$.
5. Vectors are arrows.
6. $a \cdot b$ and $a \times b$
7. Matrices (rotation, reflection, scaling)
8. Complex numbers (polar form, Euler's formula, multi-valuedness)
9. Complex vectors, Hermitian inner product, bra-ket notation
10. Vector calculus (grad, div, rot; example of point charge)
11. ODE basics
12. ODE basics
13. Probability and error analysis
14. Probability and error analysis

♣TODO♣a bit on (abstract) vector space to prep for QM?

Symbols

Mathematical Notations

Chapter 1	
$\mathbb{N}$	natural numbers
$\mathbb{N}_0$	natural numbers (0, 1, 2, ...)
$\mathbb{N}^+$	natural numbers (1, 2, ...)
$\mathbb{R}$	real numbers
$\mathbb{Q}$	rational numbers
$\mathbb{C}$	complex numbers

Greek symbols

A	α	alpha	H	η	eta	N	ν	nu	T	τ	tau
B	β	beta	Θ	θ	theta	Ξ	ξ	xi	Y	υ	upsilon
Γ	γ	gamma	I	ι	iota	O	o	omicron	Φ	$\varphi\,(\phi)$	phi
Δ	δ	delta	K	κ	kappa	Π	π	pi	X	χ	chi
E	ε	epsilon	Λ	λ	lambda	P	ρ	rho	Ψ	ψ	psi
Z	ζ	zeta	M	μ	mu	Σ	σ	sigma	Ω	ω	omega

Gray-outed ones are almost never used, but others are almost always used. You need to write them so that **others can distinguish each from others**, but Sho thinks Japanese- or Chinese-speakers are OK with it, as we everyday read and write 日 vs 曰 vs 白 or 土 vs 士 vs 工 properly!

Quiz

Q0.1 Read out the following words.

(1) α-particle
(2) β-decay
(3) Γ-function
(4) ε-δ definition
(5) angle θ and φ
(6) μ- and τ-leptons
(7) $\psi(x)$ and $\varphi(x)$
(8) ζ and ξ
(9) metric $\eta_{\mu\nu}$
(10) X-ray and γ-ray
(11) Ω-baryon
(12) $\varepsilon_{\mu\nu\rho\sigma}$-tensor

Problems

0.1 Write the following letters so that people can distinguish each from others.

(1) a, α	**(2)** b, β	**(3)** c, C	**(4)** e, E, ε
(5) ζ, ξ	**(6)** $\Theta, \theta, Q, \sigma, 6$	**(7)** $g, q, 9$	**(8)** $i, l, 1, I$
(9) k, K, κ	**(10)** m, μ	**(11)** p, ρ	**(12)** s, S
(13) t, τ, T	**(14)** u, υ, ν	**(15)** w, W, ω	**(16)** x, X, χ

SI Units

Historically, different civilizations used different units. Today, science and engineering worldwide use the **SI units** (Système International d'Unités)—the single international standard. It is built from seven **SI base units**:

Table 1: SI base units and their symbols for dimension [1]. Recall that units are case-sensitive.

quantity	symbol and name		symbol for dimension
time	s	(second)	T
length	m	(meter)	L
mass	kg	(kilogram)	M
electric current	A	(ampere)	I
temperature	K	(kelvin)	Θ
amount of substance	mol	(mole)	N
luminous intensity	cd	(candela)	J

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Bibliography

[1] Bureau International des Poids et Mesures, "The International System of Units (SI)," BIPM. [Online]. Available: https://doi.org/10.59161/AUEZ1291

Chapter 1

Derivative (Review)

1.1 The first step

Birds sing, fish swim, flowers bloom, stars twinkle, and university students calculate derivatives. Let us begin with basic **derivatives**. Try the next quiz—and note how long it takes.

Quiz

Q1.1 Calculate the first derivatives of the following functions, **measuring how many minutes it takes.**

(1) $(x+1)^3$ (2) $\tan x$ (3) $2\cos^2 x$ (4) $2\cos x^2$

(5) $x^3\cos x$ (6) $\dfrac{3\sin x}{x}$ (7) $x^{3/2}$ (8) $\sqrt{\sin x}$

- Less than two minutes? Amazing! It is as fast as Sho!
- Less than four minutes? Great, it is exactly as Sho anticipates. Please continue your effort!
- Even if you took more than four minutes, do not worry. At least you don't make any mistakes. A little more practice will help you. Try more problems in the Derivative Boot Camp.[#1]
- If you make any mistakes in these eight calculations, then it is a serious issue—just as serious as forgetting how to do the calculations at all. In university, you will perform similar calculations more than 100 times. You need to be both fast and accurate.

At university, students often underestimate the importance of basic calculations. In physics, simple calculations appear constantly, so your speed and accuracy directly affect how well you follow lectures, how efficiently you study, and ultimately your grade. Both can be improved with practice.

This course will help you strengthen these basic calculations—some of which you already know from high school—while also introducing new topics that are essential for physics. You will solve many problems and drills, just as an athlete repeats the same move hundreds of times to master it.

This document (this lecture course) is designed for first-year students in their second semester, **who have already studied derivatives well** in their first-semester calculus. As we do not cover the details of derivatives here, if you are not confident with derivatives, please first study the Derivative Boot Camp.

[#1] Visit `https://misho104.github.io/LecturePublic` and find `gp1_boot1_deriv_true.pdf`.

1.2 Typical Pitfalls

1.2.1 Notations

Consider a function $f(x)$. As we vary the value of x, x is called a **variable**. For a function $f(x)$, there are several equivalent ways to write its derivative:

$$f'(x) \quad = \quad \frac{\mathrm{d}f}{\mathrm{d}x}(x) \quad = \quad \frac{\mathrm{d}f(x)}{\mathrm{d}x} \quad = \quad \frac{\mathrm{d}}{\mathrm{d}x}f(x) \qquad \text{[All means the same thing].} \tag{1.1}$$

Similarly, the value of $f'(x)$ at a specific point $x = 3$ can be written as

$$f'(3) \quad = \quad f'(x)\Big|_{x=3} \quad = \quad \frac{\mathrm{d}f}{\mathrm{d}x}\bigg|_{x=3} \quad = \quad \frac{\mathrm{d}f}{\mathrm{d}x}(3) \quad = \quad \frac{\mathrm{d}}{\mathrm{d}x}f(3), \tag{1.2}$$

and they are all equivalent. For example, if $f(x) = 3x^2 + 6$, then $f'(x) = 6x$ and $f'(3) = 18$.

Most of students get confused when a **constant** a appears. If a is *declared as a constant*, we can define a function such as $g(x) = ax^2 + 2a$, for which

$$g'(x) = 2ax. \tag{1.3}$$

We can evaluate $g'(x)$ at $x = a$. The result is $g'(a) = 2a^2$, written as

$$g'(a) \quad = \quad g'(x)\Big|_{x=a} \quad = \quad \frac{\mathrm{d}g}{\mathrm{d}x}(a) \quad = \quad \frac{\mathrm{d}}{\mathrm{d}x}g(a) \quad = \quad 2a^2. \tag{1.4}$$

Be careful: The notation $\frac{\mathrm{d}}{\mathrm{d}x}g(a)$ does **not** mean $\frac{\mathrm{d}}{\mathrm{d}x}\big[g(a)\big] = \frac{\mathrm{d}}{\mathrm{d}x}(a^3 + 2a) = 0$.

Fail safe note: If you can't see Eq. (1.3), try setting $a = 3$. Then, you notice $g(x)$ is the same function as $f(x)$, and $g'(x)$ should equal $f'(x)$.

Quiz

Q1.2 Let a be a constant and $g(x) = ax^2 + 2a$. Calculate

(1) $g(1)$	**(2)** $g(a)$	**(3)** $g(0)$	**(4)** $g(7)$
(5) $g'(1)$	**(6)** $g'(a)$	**(7)** $g''(2)$	**(8)** $\frac{\mathrm{d}}{\mathrm{d}x}g(7)$

One more bad news. *Physicists are often lazy* and write $f(x)$ as f. If $f(x) = 2x^2 + 1$, we may write $f' = 4x$ and $f'' = 4$. For $g(t) = 2t^2 - 1$, you may write $g' = 4t$ and $g'' = 4$. Therefore, when you see a function f, you must **identify its variable from the context**.

In physics, a variable can be a function of another variable. The kinetic energy $K(v) = mv^2/2$ is a good example. Its variable v is a function of time t —we write this fact by $v = v(t)$— and thus

$$K(v) = \frac{1}{2}mv^2 \quad \text{but also} \quad K(t) = K(v(t)) = \frac{1}{2}mv(t)^2. \tag{1.5}$$

Now, what does K' mean? —It is ambiguous and we must avoid such notation. Even though, if you see it in a textbook, you need to guess the author's intention from the context.

Advanced note: This rewrite, $K(t)$ and $K(v)$, is common in physics. However, mathematicians do not like this practice because they consider K a unique object: a function should always mean the same rule applied to its input. If they see $K(v) = mv^2/2$, they would say $\left[K \text{ is an object that converts its input into } m/2 \times (\text{input})^2\text{, and thus if you feed } t \text{ into } K\text{, you must get } K(t) = mt^2/2\right]$. This is of course not what we mean. For example, imagine $v(t) = \alpha t$ with α a constant. Then

$$\text{physicists:} \quad K(v) = \frac{1}{2}mv^2, \quad K(t) = \frac{1}{2}m\alpha^2 t^2; \tag{1.6}$$

but mathematicians would say, $\left[\text{since it returns } m\alpha^2/2 \times (\text{input})^2\text{, it is a different object}\right]$ and hence use a different name for it, such as $\tilde{K}$:

$$\text{mathematicians:} \quad K\big(v(t)\big) = \frac{1}{2}m \times v(t)^2 = \tilde{K}(t) = \frac{1}{2}m\alpha^2 t^2. \tag{1.7}$$

Both conventions are reasonable, and eventually you will get used to both.

Higher-order derivatives is written as

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right) = \frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = f''(x) = f^{(2)}(x), \qquad \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right)\right) = \frac{\mathrm{d}^3 f}{\mathrm{d}x^3} = f'''(x) = f^{(3)}(x), \tag{1.8}$$

etc. Please be careful on the position of “2” and “3”.

1.2.2 Radian and trigonometric functions

At university, angles are almost always measured in **radians**:

$$360 \text{ degree} \quad (\text{or: } 360°) \quad = \quad 2\pi \text{ radian} \quad (\text{or: } 2\pi \text{ rad}) \tag{1.9}$$

Furthermore, we usually omit “radian” (because we are lazy!). So,

$$\text{a right angle is } \pi/2. \quad \text{The sum of the interior angles of a triangle is } \pi. \tag{1.10}$$

Quiz

Q1.3 Express the following angles in radians, and radians in angles.

(1) $180°$	**(2)** $45°$	**(3)** $60°$	**(4)** 90	**(5)** $-30°$
(6) $1°$	**(7)** 1	**(8)** -0.3	**(9)** $x°$	**(10)** x

Why do insist on radians? The answer comes from the derivative formula

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x. \tag{1.11}$$

Because we have chosen radians as the standard, $(\sin x)'$ becomes this simple.

Advanced note: The simpleness of Eq. (1.11) originates in the fact $\sin(0.01 \text{ rad}) \approx 0.01$. If we used degrees, we would have $\sin(0.01°) \approx 0.01 \times 0.017453$ and everything is messed up with this number 0.017453.

There are a few remarks in the notation of **trigonometric functions**:

$$\begin{aligned} &\sin^2 x \neq \sin x^2. \quad &&\text{Namely,} \quad (\sin x)^2 = \sin^2 x \quad &&\neq \quad \sin x^2 = \sin(x^2). \\ &\tan^{-1} x \neq \frac{1}{\tan x}. \quad &&\text{Namely,} \quad \tan^{-1} x = \arctan x \quad &&\neq \quad (\tan x)^{-1} = \frac{1}{\tan x} = \cot x. \end{aligned} \tag{1.12}$$

The following expressions are not incorrect but confusing;

$$\sin^{-2} x, \quad \sin^{1/2} x, \quad \sin(x)^2, \quad \sin{(x+1)}^2, \ldots$$

We should avoid ambiguity, so please *never* use confusing these notations. Sho thinks we should use $\sin^k x$ only for $k = 2, 3, 4, \ldots$, and use $\arcsin x$ instead of $\sin^{-1} x$.

1.2.3 Several interpretations of derivatives

When you, physics learners, discuss $f'(t)$, you should have the following three interpretations:

- Regarding t as the time, $f'(t)$ is the **rate of change** of $f(t)$ per unit time. If $f'(t) > 0$, the function f is increasing at the time t, while $f'(t) < 0$ means it is decreasing.
- Consider a graph of $f(t)$. Then, $f'(t)$ is the slope of the **tangent line** to the graph at the point t.
- Mathematically, $f'(t)$ is defined by (Check that they are equivalent.)

$$f'(t) := \lim_{\Delta t \to 0} \frac{f(t+\Delta t) - f(t)}{\Delta t} = \lim_{h \to 0} \frac{f(t+h) - f(t-h)}{2h} = \lim_{s \to t} \frac{f(s) - f(t)}{s - t}. \tag{1.13}$$

If you are unsure of them, please consult your first-year Calculus textbooks for further information.

Quiz

Q1.4 What is the definition of $f'(x)$? Explain.

In physics, it is important to memorize and understand the **definition** of each concept.

1.3 Taylor expansion

Reviewing the definition of $f'(x)$, we have

$$f'(x) := \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}, \qquad f'(a) := \lim_{\delta \to 0} \frac{f(a+\delta) - f(a)}{\delta}. \tag{1.14}$$

The second equation is interpreted as follows:

$$\text{If } \delta \approx 0, \quad f'(a) \approx \frac{f(a+\delta) - f(a)}{\delta}, \quad \text{i.e.,} \quad f(a+\delta) \approx f(a) + \delta\, f'(a) \tag{1.15}$$

and this interpretation is useful in the following example:

Example 1.1

Find the approximate value of the following expressions without using calculators. Then, check your answer with a calculator.

(1) $\sqrt{1.002}$ **(2)** $(1.002)^{10}$ **(3)** $\sqrt{4.004}$

Solution

(1) Apply Eq. (1.15) for $f(x) = \sqrt{x}$, $a = 1$, and $\delta = 0.002$. Then,

$$f'(x) = \frac{1}{2\sqrt{x}}, \quad f'(a) = f'(1) = \frac{1}{2}, \quad f(a+\delta) \approx f(a) + \delta f'(a) = 1 + \frac{\delta}{2} = \underline{1.001}.$$

(2) Doing the same thing for $g(x) = x^{10}$ with $a = 1$ and $\delta = 0.002$,

$$g'(x) = 10x^9, \quad g'(a) = g'(1) = 10, \quad g(a+\delta) \approx g(a) + \delta g'(a) = 1^{10} + 10\delta = \underline{1.02}.$$

(3) Doing the same thing for $h(x) = \sqrt{x}$, $a = 4$, and $\delta = 0.004$, we have

$$h'(4) = \frac{1}{4}, \quad h'(4.004) \approx h(4) + 0.004 \times \frac{1}{4} = \underline{2.001}.$$

Advanced note: "Solutions" contain not only the answer but also how you reached the answer. You are, of course, asked to write such explanations when you solve problems.

Namely, with this technique, you can find **the value of** $f(x)$ **around a point** $x = a$**.** This technique, Eq. (1.15), is a special case of Taylor's theorem, which is discusssed in ♣TODO♣???.

Quiz

Q1.5 Find the approximate value of the following expressions without using calculators. Then, check your answer with a calculator.

(1) $\sqrt{1.001}$ **(2)** $(1.0001)^{30}$ **(3)** $\dfrac{1}{1.001}$

To summarize, we have the following statement:

Statement 1.2 (Taylor expansion: basic)

For a physicists-friendly function $f(x)$,

$$f(a+\delta) \approx f(a) + \delta f'(a), \tag{1.16}$$

or as an equivalent expression,

$$f(x) \approx f(x_0) + (x - x_0) f'(x_0). \tag{1.17}$$

Quiz

Q1.6 Check these two equations are equivalent. You will be ready to use either of them in the future.

Q1.7 Review how to derive Eq. (1.16) and summarize it in your own words.

Problems

1.1 Calculate the following expression.

(1) $\sin 30°$ **(2)** $\cos 45°$ **(3)** $\cos 120°$ **(4)** $\sin 150°$ **(5)** $\tan 180°$
(6) $\cos 0$ **(7)** $\sin 2\pi$ **(8)** $\cos \pi$ **(9)** $\sin(\pi/3)$ **(10)** $\cos(-\pi/4)$
(11) $\tan(2\pi/3)$ **(12)** $\sin(\pi/6)$ **(13)** $\tan(5\pi/6)$ **(14)** $\cos(-\pi/6)$ **(15)** $\tan(-\pi/2)$

1.2 For the following $f(x)$, calculate $f'(x)$, $f''(x)$, $f^{(3)}(x)$, $f^{(4)}(x)$, and $f^{(5)}(x)$.

(1) x^3 **(2)** $\sin 2x$ **(3)** e^{2x} **(4)** $\ln 2x$

1.3 Find the approximate value for the following expressions without calculators. Then, check your answer with a calculator.

(1) $(1.002)^7$ **(2)** $(2.002)^7$ **(3)** $(0.998)^4$ **(4)** $(10.01)^3$ **(5)** $1 \div 0.998$
(6) $\sqrt{0.998}$ **(7)** $\sqrt{4.001}$ **(8)** $(0.999)^{1/2}$ **(9)** $(0.999)^{-1/2}$ **(10)** $(0.999)^{-3/2}$
(11) $\dfrac{1}{(2.02)^2}$ **(12)** $\tan 0.001$ **(13)** $\cos 0.002$ **(14)** $\sin 0.003$ **(15)** $\ln 1.001$

***1.4 Find the approximate value for the following expressions without calculators. Then, check your answer with a calculator.

(1) $\sqrt[3]{1.001}$ **(2)** $\sqrt[3]{8.012}$ **(3)** $1.001^{0.5}$ **(4)** $1.001^{-0.9}$ **(5)** $\sin(3.14)$
(6) $\cos(1.57)$ **(7)** $\sin 1°$ **(8)** $\cos 61°$ **(9)** $\ln 0.999$ **(10)** $\log_{10}(10.01)$
(11) $\sqrt{1.001}\sin 0.001$

> **Fail safe note:** If you are not confident with the meaning of $\sqrt[3]{1.001}$, $(1.001)^{-0.9}$, etc., please go to ♣TODO♣a chapter...

**1.5 For the following $f(x)$, calculate $f^{(n)}(x)$ for all positive integers $n = 1, 2, 3, \ldots$.

(1) x^{10} **(2)** $\sin 2x$ **(3)** e^{2x} **(4)** $\ln x^2$ **(5)** $\sqrt{x}$

**1.6 Write down the definition of the second derivative $f''(x)$. Repeat the discussion of Eq. (1.15) to find the expansion $f'(a+\varepsilon) \approx f'(a) + \varepsilon f'(a) + \left(\varepsilon^2/2\right) f''(a)$.

> **Advanced note:** The two equations in Eq. (1.14) have similar but different meanings: the first one defines a new function $f'(x)$, while the second one defines a number that is eventually equal to $f'(x)|_{x=a}$. Anyway, we don't care the difference.

Chapter 2

Units and Significant figures

A **physical quantity** is usually not just a number, but also has a **unit** and an **uncertainty** (error). For example, "(72 ± 2) kg" has a number "72" and a unit "kg", which compose the **central value** "72 kg". It also has uncertainty because every measurement has an uncertainty. In experimental physics, uncertainty is usually more important than the central value, and thus uncertainty handling is a fundamental skill of physicists.

In this chapter, we first review physical quantities and how to handle units. Then, we discuss **significant figures**, a simple and convenient method to express the uncertainty. Further discussions on uncertainty analysis are given in Chapter ♣TODO♣chapter.

2.1 Physical quantity

The expression "(72 ± 2) kg" means the central value is 72 kg and the **absolute uncertainty** is 2 kg. Accordingly, its **relative uncertainty** is given by $(2\,\mathrm{kg})/(72\,\mathrm{kg}) = 0.028$. Let's see other examples.

	unit	central value	absolute uncertainty	relative uncertainty
10 m ± 1 cm	m (meter)	10 m	1 cm = 0.01 m	0.001 (or 0.1%)
1.6 A ± 0.04 A	A (ampere)	1.6 A	0.04 A	0.025 (or 2.5%)
$(5 \pm 1) \times 10^{-3}$ C	C (coulomb)	0.005 C	0.001 C	0.2 (or 20%)
(50 ± 1) m/s	m/s	50 m/s	1 m/s	0.02 (or 2%)

Remark: Upper- and lowercase letters are distinguished. "M" **does not** mean "meter". The unit "A" is "ampere", not "Ampere". The unit "coulomb" is "C", not "c".

Quiz

Q2.1 For each of the following values, find its unit, central value, absolute uncertainty, and relative uncertainty.

(1) 50 kg ± 500 g **(2)** (0.05 ± 0.001) A

(3) $(1.6 \pm 0.001) \times 10^{-19}$ C **(4)** 72 km/h ± 1 m/s

Q2.2 The next passage has six (6) errors in the use of uppercase and lowercase letters. Find them out.

In the SI system, temperature is expressed in k (kelvin), a unit named after Lord Kelvin. The unit of force is N (newton), named after the british scientist Isaac newton. The units A (Ampere) and c (coulomb) are named after French scientists. In contrast, Kg (kilogram) is not named after a person. , but comes from Greek.

As an undergraduate student, you need to follow the following rules:

Statement 2.1 (Rules for Physical Quantities: Basic)

1. Numbers are always with units, even in calculations.
2. Include units in a symbol.
3. For physical quantities, use decimals (e.g., 1.6 A). Do not use fractions like (8/5) A.
4. Use significant figures to express the measurement precision.

2.2 Units

When we educate kids, we write "my height is $\boxed{}$ cm", or "my height is h cm", where $h = 172$ is just a number. But this is not nice! We want to convert the units freely and write equations such as $1.72\,\mathrm{m} = 172\,\mathrm{cm} = 0.00172\,\mathrm{km}$. So, **we always include units in symbols** $\boxed{h = 172\,\mathrm{cm}}$. Then

we can write: $\boxed{h = 172\,\mathrm{cm} = 1.72\,\mathrm{m} = 0.00172\,\mathrm{km} = 1.82\times10^{-16}\,\text{light-year.}}$

Similarly, if $m = 110\,\mathrm{g}$ and $\mathrm{g} = 9.8\,\mathrm{m/s^2}$,

we may write: $\boxed{w = mg = 110\,\mathrm{g} \times 9.8\,\mathrm{m/s^2} = 1.1\,\mathrm{kg{\cdot}m/s^2}}$

but not: $\boxed{w = mg = 0.11 \times 9.8 = 1.1\,\mathrm{kg{\cdot}m/s^2}}$

This second equation is incorrect because m is not equal to 0.11; m is equal to 0.11 kg or 110 g.

Remark: Usually, physicists use upright fonts for units and *italic fonts* for quantities. For example, m, T, and C are quantities, which can be mass, temperature, capacitance, etc. Meanwhile, m, T, and C are units: meter, tesla, and coulomb, respectively.

Quiz

Q2.3 **(1)** If $mr\omega^2 = 10.0\,\mathrm{kg{\cdot}m/s^2}$, $m = 5.0\,\mathrm{kg}$, and $r = 1.0\,\mathrm{m}$, what is ω?

(2) If $v_0 t + \frac{1}{2}at^2 = 5.0\,\mathrm{m}$, where $a = -4.0\,\mathrm{m/s^2}$ and $v_0 = 7.0\,\mathrm{m/s}$, what is t?

(3) At time $t = 0$, a particle is at $(x, y) = (2.0\,\mathrm{m}, 0)$. It moves with a constant velocity $(v_x, v_y) = (3.0, 4.0)\,\mathrm{m/s}$. What is its position at $t = 1.0\,\mathrm{s}$?

Every physical concept has its own unit. For example, speed has m/s, acceleration has $\mathrm{m/s^2}$, and energy has $\mathrm{kg{\cdot}m/s^2} = \mathrm{N{\cdot}m} = \mathrm{J}$. The table below lists the quantities you have learned. Notice that angle (rad) has no dimension. It is called a **dimensionless** quantity.

concept	dimension	typical unit	
speed	$\mathsf{L\,T^{-1}}$	m/s	
acceleration	$\mathsf{L\,T^{-2}}$	$\mathrm{m/s^2}$	
linear momentum	$\mathsf{M\,L\,T^{-1}}$	kg·m/s	
force	$\mathsf{M\,L\,T^{-2}}$	$\mathrm{kg{\cdot}m/s^2}$	(= N)

concept	dimension	typical unit	
energy / work	$\mathsf{M\,L^2\,T^{-2}}$	$\mathrm{kg{\cdot}m^2/s^2}$	$(=\mathrm{J})$
frequency	$\mathsf{T^{-1}}$	$1/\mathrm{s}$	$(=\mathrm{Hz})$
angle	1 (dimensionless)	rad	

Advanced note: To understand why angle has no dimension, you may consider the definition of the radian: it is defined as the ratio of the arc length to the radius, so the units cancel (meter / meter = 1).

Here, please do not write: The force is $\mathrm{kg{\cdot}m^2/s^2}$. This is also called newton (N)., because *we cannot mix units and concepts*. A correct (and easy) way is to use English words:

the unit of the force is $\mathrm{kg{\cdot}m/s^2}$ or the force has the unit $\mathrm{kg{\cdot}m/s^2}$.

Similarly, the following statement is incorrect because it mixes concepts and units:

Since $m = \mathrm{kg}$ and $\mathrm{g} = \mathrm{m/s^2}$, $mg = \mathrm{kg{\cdot}m/s^2} = \mathrm{N}$.

Instead, you need to write

Since m has a unit of kg and g has a unit of $\mathrm{m/s^2}$, mg has a unit of $\mathrm{kg{\cdot}m/s^2} = \mathrm{N}$.

Notice the last equation: units can be equal to other units, so we may write $\mathrm{N} = \mathrm{J/m} = \mathrm{kg{\cdot}m/s^2}$ $\mathrm{m} = \mathrm{J/N}$ $\mathrm{s} = \sqrt{\mathrm{kg{\cdot}m/N}}$. These are scientifically correct equations.

Remark: We sometimes use informal notations, such as $[F] = \mathrm{N}$ or $F \rightsquigarrow \mathrm{N}$ to express "the unit of F is N (newton)". Still, using an equal symbol $(=)$ is **not allowed**. Namely, $F = \mathrm{N}$ is always incorrect.

Quiz

Q2.4 What are the units of the following concepts? Write English sentences to explain it.

(1) speed **(2)** velocity **(3)** force **(4)** energy **(5)** power

Advanced note: In very formal situations, we use **symbols for dimensions** (see Table 1 on page 5):

Since $\dim(m) = \mathsf{M}$ and $\dim(g) = \mathsf{L\,T^{-2}}$, $\dim(mg) = \mathsf{M\,L\,T^{-2}}$,

with the operator "dim" [1]. However, it seems too complicated for most situations.

Problems

***2.1 Express the following units only with SI base units, e.g., $\mathrm{N} = \mathrm{kg}\cdot\mathrm{m/s^2}$.

(1) W (watt) (2) Pa (pascal) (3) J (joule) (4) C (coulomb)

**2.2 Use "dim" notation to express the dimensions of the following quantities. For example, if v is speed, then $\dim(v) = \mathsf{L}\,\mathsf{T}^{-1}$.

(1) speed v (2) force F

(3) angular momentum L (4) electric charge Q

(5) resistance R (6) specific heat capacity c

(7) Planck constant h (8) fine-structure constant α

**2.3 Find a few more examples of dimensionless quantities in physics.

2.3 Significant figures

In researches, we need to treat uncertainties in the method given in ♣TODO♣chap. However, most of lectures use **significant figures** to express the accuracy of a value, so that students becomes familiar with the concept of uncertainties. For example,

3.14	means the last digit "4" is uncertain. It can be 3.1***6***, 3.1***5***, 3.1***1***, 3.1***0***, ...
3.141	means "3.14" is for sure but it can be 3.14***3***, 3.14***2***, 3.14***0***, 3.1***39***, ...
3.1415	might be 3.141***6***, 3.1415***3***, etc., but the writer is sure *it is not* 3.145 or 3.135.

We write the "uncertain digit" in ***a different style*** for clarity. You should notice the difference between:

1.000***0***	meaning the last digit "0" is uncertain so it may be 1.0002, 1.0001, 0.9998, ...
1.0***0***	meaning it can be 1.03 or 0.99, but the writer is sure *it is not* 1.2 or 0.8.

It is usually convenient to use **scientific notation**:

0.00012***3***	is OK but we prefer	$1.2\boldsymbol{3}\times10^{-4}$.	Both mean the value can be $1.2\boldsymbol{2}\times10^{-4}$ etc.
0.0010***0***	is OK but we prefer	$1.0\boldsymbol{0}\times10^{-3}$.	
2997***9***	is OK but we prefer	$2.997\boldsymbol{9}\times10^{4}$.	

Fail safe note: $10^3 = 1000$ and $10^{-3} = 1/10^3 = 1 \div 1000$. Go to ♣TODO♣chap for details!

We need to *avoid ambiguous notations*. Namely,

2040	is ambiguous and not nice. We are not sure the author means 2.04×10^3 or 2.040×10^3.
120	is not nice. We are not sure the author means It may mean 1.20×10^2 or 1.2×10^2.
42000	is not nice because it has four ways to interpret the author's intention.

Quiz

Q2.5 What are the four interpretations of 42000? Write them in scientific notation.

We use **rounding**-to-the-nearest (四捨五入) when necessary. For example, if you need to convert to three significant figures, it will be

$$1.234 \to 1.23, \quad 1.235 \to 1.24, \quad 31.98 \to 32.0, \quad 1234 \to 1.23 \times 10^3, \quad \text{and so on.}$$

Quiz

★**Q2.6** Round the following numbers to three significant figures.

(1) 41.11	**(2)** 98.76	**(3)** 100.12	**(4)** 15.449
(5) 2.2360	**(6)** 0.0123456	**(7)** 9876	**(8)** 9.999×10^4

Remark: Programmers use `2.99e8` (or `2.99E8`) to mean 2.99×10^8, or `1.6e-12` (or `1.6E-12`) to mean 1.6×10^{-12}. However, we **should not** use them in handwriting.

Problems

🦾**2.4** Write the following numbers in scientific notation. However, some of them are ambiguous, so answer "ambiguous" if so.

152 340 1000 9999 43210 0.0300 0.00213 31.0 31.00 31

🦾**2.5** The expressions 0.123, 1.23, and 123 are numbers with ***three*** *significant figures*. Similarly, 1234 and 1.234×10^{-3} are numbers with ***four*** *significant figures*. How about the following expressions?

(1) 3.14	**(2)** 0.11	**(3)** 1.1×10^{-2}	**(4)** 1.1×10
(5) 1.0008	**(6)** 1.0020	**(7)** 1.0000	**(8)** 4.00×10^{-10}
(9) 0.0008	**(10)** 0.00080	**(11)** 12345	**(12)** 67890

2.4 Calculation with Significant figures

We need to do calculations of numbers with uncertainties, such as $1.23 + 4.56$ or $1.23/4.56$, but how? Here we discuss a simple method for

- addition $(x + y)$ and subtraction $(x - y)$
- multiplication $(x \times y)$ and division $(x \div y$ or $x/y)$

Other calculations, such as $\sqrt{x}$, e^x or $\sin(x)$, need the professional method given in ♣TODO♣chap.

Multiplication and Division

As a rule, if x has m significant figures and y has n significant figures, then $x \times m$ or $x \div y$ should be rounded to have $\min(m, n)$ significant figures. ...But the rule is tough to understand. Probably you should see the example, and learn through practice.

Example 2.2

Calculate the following expressions, using calculators.

(1) 8.912×2.4 **(2)** $5.0 \div 3.14$ **(3)** 2.01×49.8 **(4)** 1.210^3

Solution

Here we use a shorthand notation "3-SF" to mean "three significant figures".

(1) Since 8.912 has 4-SF and 2.4 has 2-SF, we round the result to (the smaller) 2-SF:

$$8.912 \times 2.4 = 21.3888 \to \underline{21}.$$

(2) Since 5.0 has 2-SF and 3.14 has 3-SF, we round the result to 2-SF:

$$5.0 \div 3.14 = 1.592\ldots \to \underline{1.6}.$$

(3) Both 2.01 and 49.8 have 3-SF, so we keep 3-SF, but don't forget to avoid ambiguity!

$$2.01 \times 49.8 = 100.098 \to 100 \to \underline{1.00 \times 10^2}.$$

(4) $1.210^3 = 1.210 \times 1.210 \times 1.210$, so we round the result to 4-SF:

$$1.210^3 = 1.771561 \to \underline{1.772}.$$

If you want, we can justify this treatment by carrying out a hand calculation of the long multiplication.

$$\begin{array}{ccccc}
 & & 1. & 2 & \boldsymbol{\mathit{0}} \\
\times & & 1. & 3 & \boldsymbol{\mathit{1}} \\
\hline
 & & \boldsymbol{\mathit{1}} & \boldsymbol{\mathit{2}} & \boldsymbol{\mathit{0}} \\
 & 3 & 6 & \boldsymbol{\mathit{0}} & \\
1 & 2 & \boldsymbol{\mathit{0}} & & \\
\hline
1 & 5 & \boldsymbol{\mathit{7}} & \boldsymbol{\mathit{2}} & \boldsymbol{\mathit{0}}
\end{array}
\qquad
\begin{array}{cccccc}
 & & 8. & 9 & 1 & \boldsymbol{\mathit{2}} \\
\times & & & & 2. & \boldsymbol{\mathit{4}} \\
\hline
 & \boldsymbol{\mathit{3}} & \boldsymbol{\mathit{5}} & \boldsymbol{\mathit{6}} & \boldsymbol{\mathit{4}} & \boldsymbol{\mathit{8}} \\
1 & 7 & 8 & 2 & \boldsymbol{\mathit{4}} & \\
\hline
2 & \boldsymbol{\mathit{1}} & \boldsymbol{\mathit{3}} & \boldsymbol{\mathit{8}} & \boldsymbol{\mathit{8}} & \boldsymbol{\mathit{8}}
\end{array}
\qquad
\begin{array}{cccccc}
 & & & 2. & 0 & \boldsymbol{\mathit{1}} \\
\times & & & 4 & 9. & \boldsymbol{\mathit{8}} \\
\hline
 & & \boldsymbol{\mathit{1}} & \boldsymbol{\mathit{6}} & \boldsymbol{\mathit{0}} & \boldsymbol{\mathit{8}} \\
 & 1 & 8 & 0 & \boldsymbol{\mathit{9}} & \\
 & 8 & 0 & \boldsymbol{\mathit{4}} & & \\
\hline
1 & 0 & \boldsymbol{\mathit{0.}} & \boldsymbol{\mathit{0}} & \boldsymbol{\mathit{9}} & \boldsymbol{\mathit{8}}
\end{array}$$

Observe which digits are uncertain, and how they "pollute" the results in each step. In $1.2\boldsymbol{\mathit{0}} \times 1.3\boldsymbol{\mathit{1}}$, the last $\boldsymbol{\mathit{1}}$ pollutes the part of $\boldsymbol{\mathit{1}} + 6$ and we get an uncertain number $\boldsymbol{\mathit{7}}$. So, the result is $1.5\boldsymbol{\mathit{7}}$.

Addition and subtraction

Different rules are applied for addition and subtraction. You can understand the rules easily if you do long addition and observe which digits are polluted.

Example 2.3

Calculate the following expressions.

(1) $12.3 + 4.56$ **(2)** $123 + 4.56$ **(3)** $0.50 - 0.032$ **(4)** $1.20 \times 10^3 - 27$

Solution

$$\begin{array}{r} 12.\mathbf{3} \\ +\ \ 4.5\mathbf{6} \\ \hline 16.\mathbf{86} \end{array} \qquad \begin{array}{r} 12\mathbf{3}. \\ +\ \ 4.5\mathbf{6} \\ \hline 12\mathbf{7.56} \end{array} \qquad \begin{array}{r} 0.5\mathbf{0} \\ -\ 0.03\mathbf{2} \\ \hline 0.4\mathbf{68} \end{array} \qquad \begin{array}{r} 12\mathbf{00} \\ -\ \ \ 2\mathbf{7} \\ \hline 11\mathbf{73} \end{array}$$

and we round the results. So, the answers are 16.**9**, 12**8**, 0.4**7**, and 1.1**7**$\times 10^3$.

Problems

2.6 Calculate the following, taking care of significant figures. You may use calculators.

(1) 1.23×4.5 **(2)** 1.50×2.0 **(3)** 2.00×5.00 **(4)** 4.0×2.5

(5) $1.23 + 4.5$ **(6)** $1.50 + 2.0$ **(7)** $2.00 + 8.00$ **(8)** $100.0 + 0.123$

(9) $1.23 \div 4.5$ **(10)** $6.0 \div 3.00$ **(11)** $9.00 \div 4.5$ **(12)** $1.50 \div 3.0$

(13) $1.23 - 4.5$ **(14)** $3.10 - 0.10$ **(15)** $3.10 - 0.1$ **(16)** $3.10 - 3.0$

(17) $131 + 69$ **(18)** $131 + 6.9$ **(19)** $131 + 0.69$ **(20)** $1.3 \times 10^2 + 69$

(21) $(1.2 \times 10^5) \times 9.4$ **(22)** $1.2 \div (5.2 \times 10^5)$

(23) $(3.33 \times 10^5) \times (6.3 \times 10^4)$ **(24)** $(3.33 \times 10^5) \times (6.3 \times 10^{-4})$

(25) $(3.33 \times 10^5) \div (6.3 \times 10^4)$ **(26)** $(3.33 \times 10^5) \div (6.3 \times 10^{-4})$

(27) $1.00 \times 10^4 + 2.5 \times 10^3$ **(28)** $3.00 \times 10^3 - 4.50 \times 10^2$

(29) $2.99 \times 10^2 + 0.814$ **(30)** $1.23 \times 10^{-4} + 4.5 \times 10^{-6}$

***2.7** Calculate the following, taking care of significant figures. You may use calculators.

(1) $1.50 \times 2.000 \times 3.50$ **(2)** $8.00 \div 4.00 \div 0.25$ **(3)** 5.0^3 **(4)** 2.10^3

(5) $1.0 + 2.0 + 3.0 \times 4.0$ **(6)** $17 - 8.0 - 3.0$ **(7)** $3.0^3 + 1.2$ **(8)** $5.0^3 - 5$

(9) $(1.2 + 3.45) \times 2.1$ **(10)** $(8.0 - 1.25) \div 2.0$ **(11)** $1.2 \times 3.4 + 5.6$

(12) $18.0 - 2.5 \times 1.2$ **(13)** $1.20 \times 3.40 + 5.60$ **(14)** $\left(5.0 \times 10^2 + 3.4\right) \times 1.2$

2.8 Calculate the following with calculators, taking care of significant figures and units.

(1) $3.1\,\mathrm{m/s} \times 1.0\,\mathrm{hour}$ **(2)** $3.1\,\mathrm{m} \div 25\,\mathrm{cm}$

(3) $1.5\,\mathrm{kg} \times 9.8\,\mathrm{m/s^2}$ **(4)** $1.50\,\mathrm{km} + 195\,\mathrm{m}$

(5) $\left(5.2\,\mathrm{kg/m^3}\right) \times (4.0\,\mu\mathrm{m})^3$ **(6)** $\left(5.2\,\mathrm{kg/m^3}\right) \times \left(4.0\,\mu\mathrm{m}^3\right)$

(7) $1.2\,\mu\mathrm{m} \div 2.00\,\mathrm{nm}$ **(8)** $\left(1.7 \times 10^{-27}\,\mathrm{kg}\right) \times \left(3.00 \times 10^8\,\mathrm{m/s}\right)^2$

(9) $2.000\,\mathrm{kg} \times \left(3.0\,\mathrm{m/s^2}\right)$ **(10)** $2.000\,\mathrm{kg} \times (3.0\,\mathrm{m/s})^2$

(11) $\left(1.60 \times 10^{-19}\,\mathrm{C}\right) \div 2.0\,\mu\mathrm{s}$ **(12)** $\left(1.60 \times 10^{-19}\,\mathrm{C}\right) \times 5.0\,\mathrm{kV}$

(13) $\left(6.63 \times 10^{-34}\,\mathrm{J \cdot s}\right) \times \left(3.0 \times 10^{12}\,\mathrm{Hz}\right)$ **(14)** $1.50\,\mathrm{MJ} \div 20.0\,\mathrm{m}$

Advanced note: Uncertainties can often be determined subjectively, but central values are also without specific definitions; it is often the averaged value of the measurements, but one may assume some probabilistic distribution and use its median, mean, or mode as the central value.

Chapter 3

Logic

Solve Quiz 2.3 (on page 14) again.

(1) If $mr\omega^2 = 10.0\,\mathrm{kg{\cdot}m/s^2}$, $m = 5.0\,\mathrm{kg}$, and $r = 1.0\,\mathrm{m}$, what is ω?
(2) If $v_0 t + \frac{1}{2}at^2 = 5.0\,\mathrm{m}$, where $a = -4.0\,\mathrm{m/s^2}$ and $v_0 = 7.0\,\mathrm{m/s}$, what is t?
(3) At time $t = 0$, a particle is at $(x, y) = (2.0\,\mathrm{m}, 0)$. It moves with a constant velocity $(v_x, v_y) = (3.0, 4.0)\,\mathrm{m/s}$. What is its position at $t = 1.0\,\mathrm{s}$?

The answers are

(1) $\omega = \pm 1.4\,\mathrm{s^{-1}}$ **(2)** $t = 1.0\,\mathrm{s}, 2.5\,\mathrm{s}$ **(3)** $x = 5.0\,\mathrm{m}, y = 4.0\,\mathrm{m}$

but **what do they mean?**

Quiz

Q3.1 Guess the correct meaning of each of above expressions.

(1) ω is both $+1.4\,\mathrm{s^{-1}}$ and $-1.4\,\mathrm{s^{-1}}$ v.s. ω is either $+1.4\,\mathrm{s^{-1}}$ or $-1.4\,\mathrm{s^{-1}}$
(2) t is both $1.0\,\mathrm{s}$ and $2.5\,\mathrm{s}$ v.s. t is either $1.0\,\mathrm{s}$ or $2.5\,\mathrm{s}$
(3) x is $5.0\,\mathrm{m}$ and y is $4.0\,\mathrm{m}$ v.s. Either x is $5.0\,\mathrm{m}$, or y is $4.0\,\mathrm{m}$.

Elementary educations do not discuss these differences because kids do not know **logical thinking**. Now, you are a grown-up university student. You need to think about it.

3.1 Basic Logics

Since elementary school, you have written many equalities and inequalities, such as

$$3 + 5 = 8, \quad 6 + 2 = 10, \quad 1 + 3 > -3, \quad 1 + 3 \neq 0, \quad 3^3 = 9, \quad 1 > 2, \quad \text{and} \quad \sin(\pi) \neq 0.$$

Quiz

Q3.2 An equality can be **true** (T) or **false** (F). For each of the above statements, state whether it is true or false.

Q3.3 Confirm that the following statements are all true:

(1) $3 + 5 = 8$ is true. **(2)** $6 + 2 = 10$ is false.
(3) $3 \neq 3$ is false. **(4)** False is false.
(5) "False is false" is true. **(6)** "False is true" is false.

We can only write true things, such as "$3 + 5 = 8$", "$3^2 = 9$ is true", or "$6 + 2 = 10$ is false". In other words, **when you write something, you must confirm that it is true.** Sometimes, you are asked to confirm a statement, or **prove** a statement. The process is called **proof** of the statement.

Numbers can be manipulated by operators such as $+$, $\div$. Similarly, true (T) and false (F) can be manipulated by **and**, **or**, and **not** operators.

and	(conjunction)	"A AND B" ($A \wedge B$) is true if both A and B are true, and false otherwise.
or	(disjunction)	"A OR B" ($A \vee B$) is true if at least one of A and B is true. False if both are false.
not	(negation)	"NOT A" ($\neg A$) is true if A is false, and false if A is true.

We can write these property as equations:

$$\begin{array}{ll} (\mathrm{T} \wedge \mathrm{T}) = \mathrm{T} & (\mathrm{T} \wedge \mathrm{F}) = (\mathrm{F} \wedge \mathrm{T}) = (\mathrm{F} \wedge \mathrm{F}) = \mathrm{F} \\ (\mathrm{F} \vee \mathrm{F}) = \mathrm{F} & (\mathrm{T} \vee \mathrm{T}) = (\mathrm{T} \vee \mathrm{F}) = (\mathrm{F} \vee \mathrm{T}) = \mathrm{T} \\ (\neg\, \mathrm{T}) = \mathrm{F} & (\neg\, \mathrm{F}) = \mathrm{T}. \end{array}$$

but the **truth table**, shown to the right, is more useful.

A	B	A and B	A or B	not A
T	T	T	T	F
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

Remark: Note that "$1 + 1 = 2$ or $2 + 2 = 4$" is true, which says the word "or" is a bit different from everyday English. In daily conversation, *"Sho will eat ramen or sushi tonight"* usually means "but not both". However, in mathematics, if Sho eats both ramen and sushi for a dinner, the statement *"Sho had sushi or ramen tonight"* is true.

Quiz

Q3.4 State whether it is true or false for the following statements.

(1) $1 + 2 = 3$ and $3 + 4 < 5$.

(2) $10 \div 3 > 0$ or $10 \div 2 > 0$.

(3) not $(3 > 5)$.

(4) $6 + 3 > 0$ and not $(5 - 2 > 0)$.

3.2 Implication $\Longrightarrow$

We can discuss the following statements: are they true or false?

Example 3.1

- If $x > 5$, then $x > 1$. ... This is true.
- If $x < 9$, then $x < 2$. ... This is false, because we have a counterexample $x = 5$.
- If $x^2 < 1$, then $x < 1$. ... This is true.
- If $x^2 = 1$, then $x = -1$. ... This is false, because we have a counterexample $x = 1$.

In general, it is easy to claim that a statement is false, i.e., to *refute* or *disprove* a statement. You just have to find one **counterexample**. Meanwhile, it is more difficult to claim that a statement is true. As mentioned above, you need to write a proof.

Now, let's try to find some counterexamples.

Quiz

Q3.5 The following statements are false. Find a counter example for each.

(1) If $x^2 > 0$, then $x > 0$. **(2)** If $x^2 - x = 0$, then $x = 0$.

(3) If $x^2 = y^2$, then $x = y$. **(4)** If $x + y < 1$, then $x < 1$ and $y < 1$.

We use the symbol "$\Rightarrow$" to express "if ... then ..." statement.

$A \Longrightarrow B$ means If A is true, then B is true. or equivalently, A implies B.

and this is called **implication**. For example, the statements in the previous quiz, which you already have found counterexamples, can be written as

$$x^2 > 0 \Rightarrow x > 0, \quad x^2 - x = 0 \Rightarrow x = 0, \quad x^2 = y^2 \Rightarrow x = y, \quad x + y < 1 \Rightarrow x < 1 \land y < 1.$$

Quiz

Q3.6 Similarly rewrite the statements in Example 3.1 with using the symbol "$\Longrightarrow$".

Advanced note: In formal logic, $A \Rightarrow B$ is defined by $(\neg A) \lor B$, which means "if A is true, B must also be true; but if A is false, we do not care about B". So, if A is always false, $A \Rightarrow B$ is always true, no matter what B is. For example, $(x \in \mathbb{R} \land x^2 < -1) \Rightarrow x = 999$ is a true statement. Through this definition, it is clear that $A \Rightarrow B$ is equivalent to $\neg B \Rightarrow \neg A$, which we will see below.

3.3 Equivalent Statements—What does "solve" mean?

Since elementary school, you have solved many equations, but what does "solving an equation" mean?

Quiz

Q3.7 **(1)** Choose the true statements.

(a) $2x - 1 = 0 \Longrightarrow x = 1/2$. **(b)** $x^2 = 4 \Longrightarrow x = 2$.

(c) $2x - 1 = 0 \Longrightarrow x > -10$. **(d)** $x^2 = 4 \Longrightarrow x = 0$.

(2) Do you think these statements can be considered as "solving an equation"?

(b) and (d) are false statements (find counterexamples!), so we cannot write them. (a) and (c) are true statements, and (a) looks "solving an equation", but we do not consider (c) is "solving an equation". The difference between (a) and (c) is that

$$2x - 1 = 0 \Longleftarrow x = 1/2 \quad \text{is true, but} \quad 2x - 1 = 0 \Longleftarrow x > -10 \quad \text{is false.}$$

So, for (a), both $\Longrightarrow$ and $\Longleftarrow$ are correct: we use $\Longleftrightarrow$ to express such cases.

Definition 3.2 (Equivalent)

If both $A \Longrightarrow B$ and $A \Longleftarrow B$ are true, we call "A and B are **equivalent**", and write $A \Longleftrightarrow B$.

$A \Rightarrow B$ is read by "B if A" in English. Similarly, $A \Leftrightarrow B$ is read by "B if and only if A" or "B **iff** A". (namely, **if**-and-only-**if**).

Now you need to do a bit of practice. It is actually tough, unfortunately, but this drill is important for logical thinking.

Problems

3.1 Fill in the blanks with $\Longrightarrow$, $\Longleftarrow$, or $\Longleftrightarrow$, where x is a real number.

(1) $x = 2 \;\boxed{}\; x > 0$	**(2)** $x > 1 \;\boxed{}\; x > 0$	**(3)** $x = 1 \;\boxed{}\; x > 0$
(4) $x = -3 \;\boxed{}\; x^2 = 9$	**(5)** $x = 0 \;\boxed{}\; x^2 = 0$	**(6)** $x < 0 \;\boxed{}\; x^2 > 0$
(7) $x = 3 \;\boxed{}\; x \neq 5$	**(8)** $3x = 3 \;\boxed{}\; x = 1$	**(9)** $x^3 = x^2 \;\boxed{}\; x = 1$

If the statement "$A \Rightarrow B$" is true, then

- B is called a **necessary condition** for A, because B is necessary for A; if not B, then not A.
- A is called a **sufficient condition** for B, because if A is true, B is "sufficiently" true.

So, if A and B are equivalent, B (resp. A) is called *necessary-and-sufficient condition* for A (resp. B). For example, because $x = 1 \Longrightarrow x^2 = 1$,

- $x = 1$ is sufficient to satisfy $x^2 = 1$. ($x = 1$ is a sufficient condition for $x^2 = 1$.)
- $x^2 = 1$ is necessary to satisfy $x = 1$. ($x^2 = 1$ is a necessary condition for $x = 1$.)

Quiz

Q3.8 Fill in the blanks with "a sufficient condition" or "a necessary condition".

(1) $x = -1$ is $\boxed{}$ for $x^2 = 1$.	**(2)** $x^2 = 4$ is $\boxed{}$ for $x = 2$.
(3) $x > 1$ is $\boxed{}$ for $x > 0$.	**(4)** $x^2 > 0$ is $\boxed{}$ for $x < 0$.
(5) $x < 0$ is $\boxed{}$ for $x^2 > 0$.	**(6)** $x^2 > 0$ is $\boxed{}$ for $x > 0$.

"To solve an equation" means "to find an *equivalent* equation in the form of $x = \boxed{}$". Since $2x - 1 = 0 \Longleftrightarrow x = 1/2$ and $x^2 = 4 \Longleftrightarrow x = \pm 2$, we say $x = 1/2$ and $x = \pm 2$ are the solutions of the equations, respectively.

When you solve an equation, you have to check that the solution is **necessary and sufficient**.

- If you don't check necessity, you may have an "incomplete solution".
- If you don't check sufficiency, you may have an "extraneous solution".

Quiz

Q3.9 **(1)** Solve $\sqrt{x+2} = x$. You might reach an *extraneous solution*, which you need to *exclude* it.

(2) Solve $\sqrt{x^2} = 4$. You might reach an *incomplete solution*, where you need to find more solutions.

> **Fail safe note:** Recall that $\sqrt{x^2} = x$ is incorrect. (What should it be?)

Before discussing advanced topics on logics, you should do some drills.

Problems

3.2 State whether each statement is true (T) or false (F). For example, $2 > 5$ is "F".

(1) $(3 > 1) \wedge (2 < 5)$ **(2)** $3 < 1 \vee 2 > 5$ **(3)** $\neg(3 > 1)$

(4) $\neg(2 > 5)$ **(5)** $(\neg(3 < 1)) \wedge (2 < 5)$ **(6)** $\neg(3 < 1 \wedge 2 < 5)$

(7) $(4 = 4) \wedge \neg(2 > 5)$ **(8)** $(3 < 1) \vee \neg(3 = 1)$ **(9)** $(\neg(3 < 1)) \vee \neg(3 > 1)$

3.3 State whether each statement is true or false. If false, give a counter example.

(1) If $x = 2$, then $x^2 = 4$. **(2)** If $x^2 = 4$, then $x = 2$.

(3) $x^2 = 0 \implies x = 0$ **(4)** $x > 2 \implies x > 0$

(5) $x > 0 \implies x > 2$ **(6)** $x = 3 \implies |x| = 3$

(7) $(x = 1) \vee (x = -1) \implies x^2 = 1$ **(8)** $x^2 = 1 \implies (x = 1) \vee (x = -1)$

★3.4 Solve each equation. State the complete solution set, including all real solutions. If there is no real solution, say so.

(1) $\sqrt{x} = 3$ **(2)** $\sqrt{x-1} = 2$ **(3)** $|x| = 5$ **(4)** $|x| = |-3|$

(5) $\sqrt{x^2} = 3$ **(6)** $\sqrt{(x-1)^2} = 2$ **(7)** $|x^2| = 5$ **(8)** $|x-2| = 3$

(9) $|2x+1| = 7$ **(10)** $|x+4| = 0$ **(11)** $x^2 + 1 = 0$ **(12)** $x^6 = x^4$

*****3.5** Let x and y be real numbers. Fill in the blanks with $\implies$, $\impliedby$, or $\iff$. If your answer is $\implies$ or $\impliedby$, explain the reason by giving relevant counter examples.

(1) $x = 2 \;\boxed{\quad}\; x^2 = 4$ **(2)** $x^2 = 4 \;\boxed{\quad}\; x = 2 \vee x = -2$

(3) $x > 0 \;\boxed{\quad}\; x^2 > 0$ **(4)** $|x| = 2 \;\boxed{\quad}\; (x = 2 \vee x = -2)$

(5) $x + 1 = 0 \;\boxed{\quad}\; x = -1$ **(6)** $x^2 = x \;\boxed{\quad}\; x(x-1) = 0 \;\boxed{\quad}\; x = 0$

(7) $\sqrt{x} = 123 \;\boxed{\quad}\; x = 123^2$ **(8)** $\sqrt{x^2} = 0 \;\boxed{\quad}\; x = 0$

(9) $\sqrt{x^2} = 3 \;\boxed{\quad}\; x = 3$ **(10)** $\sqrt{x^2} = 3 \wedge x > 0 \;\boxed{\quad}\; x = 3$

(11) $\sqrt{x^2} = \sqrt{y^2} \;\boxed{\quad}\; x = y$ **(12)** $x + y > 0 \;\boxed{\quad}\; x > 0 \wedge y > 0$

(13) $xy = 0 \;\boxed{\quad}\; x = y = 0$ **(14)** $x^2 + y^2 = 0 \;\boxed{\quad}\; x = y = 0$

**3.6 Let a be a real constant and x be a real number. For (1)–(4), fill in the blanks with $\Longrightarrow$, $\Longleftarrow$, or $\Longleftrightarrow$. If your answer is $\Longrightarrow$ or $\Longleftarrow$, explain the reason by counter examples.

(1) $\sqrt{x} = 4$ $\boxed{}$ $x = 16$ (2) $\sqrt{x^2} = 4$ $\boxed{}$ $x = 4$

(3) $\sqrt{x} = a^2$ $\boxed{}$ $x = a^4$ (4) $\sqrt{x^2} = a^2$ $\boxed{}$ $x = a^2$

(5) Solve the equation $\sqrt{x^2} = a$, noting that a can be negative.

★3.7 The following "solutions" have logical errors. Identify errors and find the right answer.

(1) A student solves $x^2 = 9$ and writes $x = 3$.

(2) He solves $x^2 - 3x = 0$ by dividing both sides by x, obtaining $x - 3 = 0$, so $x = 3$.

(3) He solves $x^6 = x^4$ by dividing both sides by x^4, obtaining $x^2 = 1$, so $x = \pm 1$.

Next, fill in the blanks with $\Longrightarrow$, $\Longleftarrow$, or $\Longleftrightarrow$.

(4) $x^2 = 9$ $\boxed{}$ $x = 3$ (5) $x^2 - 3x = 0$ $\boxed{}$ $x - 3 = 0$ $\boxed{}$ $x = 3$

(6) $x^6 = x^4$ $\boxed{}$ $x^2 = 1$ $\boxed{}$ $x = \pm 1$

***3.8 (1) Recall that $A \Rightarrow B$ is defined by $(\neg A) \vee B$. Also, recall that $A \Leftrightarrow B$ is defined by $A \Rightarrow B \wedge A \Leftarrow B$. Write a truth table (see page 22) for $A \Rightarrow B$, $A \Leftarrow B$, and $A \Leftrightarrow B$.

(2) Explain the reason we can understand $A \Leftrightarrow B$ as $A = B$.

(3) Write a truth table for the following expressions:

$$A \wedge B \quad \neg(A \wedge B) \quad (\neg A) \vee (\neg B) \quad \neg(A \vee B) \quad (\neg A) \wedge (\neg B)$$

Explain that this truth table is considered as a *proof* of **de Morgan's theorem**

$$\neg(A \vee B) = (\neg A) \wedge (\neg B), \qquad \neg(A \wedge B) = (\neg A) \vee (\neg B). \tag{3.1}$$

(4) Prove the following, which we will discuss in the next section.

$$\big(A \Rightarrow B\big) = \Big((\neg B) \Rightarrow (\neg A)\Big), \qquad \big(A \Rightarrow B\big) = \neg\big(A \wedge \neg B\big). \tag{3.2}$$

3.9 This lecture does not cover **quantifiers "for-all $\forall$" and "exists $\exists$". Learn them by yourselves. In particular, prove the following:

(1) $\exists x, P(x) \Longleftrightarrow \neg\big(\forall x, \neg P(x)\big)$ (2) $\neg\big(\exists x, P(x)\big) \Longleftrightarrow \forall x, \neg P(x)$

(3) $\forall x, P(x) \Longleftrightarrow \neg\big(\exists x, \neg P(x)\big)$ (4) $\neg\big(\forall x, P(x)\big) \Longleftrightarrow \exists x, \neg P(x)$

**3.10 For each statement, write its negation, such as "not ($x \geq 0$ for all real x)", in a natural form. Then, state whether the original statement and the negated one are true or false.

(1) $x^2 \geq 0$ for all real x. (2) There exists a real x such that $x^2 = -1$.

(3) For all real x, $x > 0 \Rightarrow x^2 > 0$. (4) $|x| = x$ for all real x.

3.4 A few more notes about logic and proof

There is a useful theorem for "not" operator, called **de Morgan's theorem**.

Theorem 3.3 (De Morgan's theorem)

1. “not (A and B)” is equivalent to “(not A) or (not B)”, i.e., $\neg(A \wedge B) = (\neg A) \vee (\neg B)$.
2. “not (A or B)” is equivalent to “(not A) and (not B)”, i.e., $\neg(A \vee B) = (\neg A) \wedge (\neg B)$.

The proof is given as Problem 3.8.

Quiz

Q3.10 Simplify the following statements by using de Morgan’s theorem.

(1) not ((not A) or (not B)) **(2)** not ((not A) and (not B))

(3) $\neg((\neg A) \wedge (\neg B))$ **(4)** $\neg(A \vee \neg B)$

Q3.11 For each of the following, write its negation (“not A”).

(1) $x > 0$ or $x < 1$. **(2)** $x \neq 4$ and $x \neq 5$. **(3)** $0 < x < 1$.

(4) $x > 0$ or $y > 0$. **(5)** $x = 0$ and $y = 0$. **(6)** $x \neq 0$ and $y \neq 0$.

In general, $A \Rightarrow B$ and $B \Rightarrow A$ are different. However, if $A \Rightarrow B$, then $\neg B \Rightarrow \neg A$ (not $B \Rightarrow$ not A) is always correct. It is called the **contrapositive** of $A \Rightarrow B$. Let’s see an example.

Example 3.4

If $0 < x < 2$, then x^2 is always less than 4. We can write it by $(0 < x < 2) \Rightarrow \left(x^2 < 4\right)$. Let A be the statement “$0 < x < 2$” and B be the statement “$x^2 < 4$”. “Not A” is “$x \le 0$ or $x \ge 2$”. Meanwhile, “not B” is “$x^2 \ge 4$”.

- Its **converse** ($A \Leftarrow B$) is $\left(x^2 < 4\right) \Rightarrow (0 < x < 2)$, which is false (counterexample: $x = -1$).
- Its **inverse** (not $A \Rightarrow$ not B) is $(x \le 0$ or $x \ge 2) \Rightarrow \left(x^2 \ge 4\right)$, which is false.
- Its **contrapositive** (not $A \Leftarrow$ not B) is $\left(x^2 \ge 4\right) \Rightarrow (x \le 0$ or $x \ge 2)$, which is true.

Because the contrapositive of a true statement is always true, we can prove a statement by proving its contrapositive. This method is called **proof by contrapositive**.

Quiz

Q3.12 For each of the following statements, write its converse, inverse, and contrapositive. Then state whether each of them is true or false.

(1) If $x > 3$, then $x > 0$. **(2)** If $x = 2$, then $x^2 = 4$.

(3) If $|x| = 0$, then $x = 0$. **(4)** If $x^2 = 4$, then $x \neq 3$.

Q3.13 Prove the following statements by *proof by contrapositive*. Namely, first write the contrapositive, then prove it.

(1) If $x^2 > 1$, then $|x| > 1$. **(2)** If $x^3 + x^2 + x < 0$, then $x < 0$.

(3) If $xy \neq 0$, then $x \neq 0$. **(4)** If n^2 is not integer, then n is not integer.

Another powerful method is **proof by contradiction** (deductio ad absurdum). Imagine you prove "if A, then B". You assume "A" and "not B" are both true, and derive a contradiction. If you reach a contradiction, it means "A and not B" is false, i.e., "not (A and not B)" is true, which means $A \Rightarrow B$.

Quiz

Q3.14 Prove the following statements by *proof by contradiction*.

(1) If x^2 is an even integer, then x cannot be an odd integer.

(2) If a, b, and c are integers and abc is even, at least one of a, b, or c is even.

Remark: The validity of "proof by contrapositive" and "proof by contradiction" was already discussed on page 26, Eq. (3.2).

Advanced note: You learned another useful method for proof, **mathematical induction**, in high school. In university, you may learn more advanced methods, such as **infinite descent** and **transfinite induction**.

3.5 Facts, Definitions, Assumptions, and Conclusions

One of the main reasons students find university physics difficult is that they treat all statements in the same way. In physics, every statement has its own role:

- experimental fact
- definition (in mathematics: definition)
- assumption / approximation (in mathematics: axiom or assumption)
- derived statements (in mathematics: theorem, corollary, or proposition)

You should always know which role each statement plays.

Remark: Mathematics does not have experimental facts because it is not based on experiments; it is purely based on definitions.

Example 3.5 (Types of statements in a calculation)

A particle moves as $x(t) = kt^2 + x_0$, where k and x_0 are constants. Find the averaged velocity between $t = 0$ and $t = t_0$.

Solution

We should first clarify assumptions and definitions:

- Assumptions (explicitly written): $x(t) = kt^2 + x_0$. k and x_0 are constants.
- Assumptions (implicit): k, x_0, and t_0 are real numbers, and t means time.
- Definition: averaged velocity is *defined* by $v_{\text{avg}} := \dfrac{x(t_0) - x(0)}{t_0 - 0}$.

Now, we can calculate $\dfrac{x(t_0) - x(0)}{t_0 - 0} = \dfrac{\left(kt_0^2 + x_0\right) - x_0}{t_0} = kt_0$ to give a derived statement:

- Derived statement (conclusion): the averaged velocity between $t = 0$ and t_0 is $v_{\text{avg}} = kt_0$.

Quiz

Q3.15 Carry out a similar discussion for "find the instantaneous velocity at $t = t_1$".

Consider an **experimental fact**, such as "gravity is always attractive". We have to accept it. We need to remember it and understand what it says. "Protons have charge $+|e|$ and electrons have charge $-|e|$" is another experimental fact, but we can understand it as the definition of the elementary charge $|e|$.

Definitions often appear as mathematical equations, such as "instantaneous velocity is defined by $v(t) \coloneqq \mathrm{d}x/\mathrm{d}t$". Another example is "uniform circular motion is a motion with constant speed along a circle"; here, the word "uniform circular motion" is defined.

Meanwhile, **assumption** is what we can impose freely. For example, you can decide whether you ignore air resistance or not. However, every conclusion is valid only under its assumptions. If you impose bad assumptions, your conclusion will be meaningless. For example, if you want to analyze the free fall of a sheet of paper but you ignore air resistance, your result should be invalid and meaningless. You must be very careful when you make assumptions.

Remark: We often use "$A \coloneqq B$" to say "define A by B". For example, $v(t) \coloneqq \mathrm{d}x/\mathrm{d}t$ is the definition of instantaneous velocity, and $T \coloneqq 1/f$ is the definition of period. Meanwhile, "$A = B$" just means "A is equal to B".

Some physicists use $A \equiv B$ to say "A is defined by B", but Sho does not recommend it because mathematicians use the symbol $\equiv$ for other purposes.

Problems

***3.11 For each of the following statements, write its converse, inverse, and contrapositive. State T or F for each of the four statements.

(1) "If $x^2 - 5x + 6 = 0$, then $x = 2$ or $x = 3$."

(2) "If $x > 0$ and $y > 0$, then $x + y > 0$."

(3) "If x and y are both even integers, then xy is divisible by 4."

***3.12 The following four statements are related as original / converse / inverse / contrapositive of one another. Identify which is which, and state T or F for each.

(1) If n^2 is divisible by 3, then n is divisible by 3.

(2) If n is not divisible by 3, then n^2 is not divisible by 3.

(3) If n is divisible by 3, then n^2 is divisible by 3.

(4) If n^2 is not divisible by 3, then n is not divisible by 3.

***3.13 Each of the following is a claimed proof by contrapositive. Check whether the contrapositive was written correctly, and whether the proof is valid.

(1) Claim: "If $x^2 + y^2 = 0$, then $x = 0$."
Proof attempt: "Suppose $x \neq 0$. Then $x^2 > 0$, so $x^2 + y^2 > 0$. Contradiction."

(2) Claim: "If n is odd, then $n^2 + n$ is even."
Proof attempt (by contrapositive): "Suppose $n^2 + n$ is odd. Then $n(n+1)$ is odd, so both n and $n+1$ are odd. But consecutive integers cannot both be odd. Contradiction." (Is this actually a proof by contrapositive or by contradiction?)

**3.14 The statement "$A \Rightarrow B$" and its contrapositive "not $B \Rightarrow$ not A" are logically equivalent. Use this fact to prove:
"If $ab = 0$, then $a = 0$ or $b = 0$."
by proving its contrapositive: "If $a \neq 0$ and $b \neq 0$, then $ab \neq 0$."

**3.15 A student claims: "I proved $\neg B \Rightarrow \neg A$, so I have proved $B \Rightarrow A$." Is this correct? Explain, and state what the student has actually proved.

***3.16 For each situation, identify the **logical error** and correct it.

(1) From $x^2 = 4$ a student writes: "squaring both sides of $x = 2$ gives $x^2 = 4$, so the solution is $x = 2$."

(2) From $\sin\theta = 1/2$ a student writes: "therefore $\theta = \pi/6$."

(3) A student solves $\sqrt{x+1} = x - 1$ by squaring: $x + 1 = (x-1)^2$, then finds $x = 0$ or $x = 3$, and concludes both are solutions.

***3.17 In the following argument, label every sentence as **experimental fact**, **definition**, **assumption**, or **derived statement**. Then identify if any step has a logical gap.

> The speed of light in vacuum is $c = 3.0 \times 10^8$ m/s. Let $n \coloneqq c/v$, where v is the speed of light in a medium. Assume the medium is glass with $n = 1.5$. Then $v = c/n = 2.0 \times 10^8$ m/s. Therefore light travels slower in glass than in vacuum.

***3.18 The following two statements look similar but are logically different. For each, state whether it is T or F and explain.

(1) For all real x: if $x > 0$, then $x^2 > 0$.

(2) For all real x: if $x^2 > 0$, then $x > 0$.

**3.19 Using De Morgan's theorem, rewrite each negation without using "not" on a compound statement.

(1) not ($x > 0$ and $y > 0$)

(2) not ($x = 0$ or $y = 0$)

(3) not (if $x > 1$, then $x^2 > 1$) (Hint: $A \Rightarrow B$ is equivalent to $(\neg A) \vee B$.)

**3.20 Find the logical error in the following "proof" that $1 = 2$.

> Let $a = b = 1$. Then $a^2 = ab$, so $a^2 - b^2 = ab - b^2$, so $(a-b)(a+b) = b(a-b)$, so $a + b = b$, so $2 = 1$.

***3.21 True or false? Give a reason.

(1) The contrapositive of a false statement is also false.

(2) If $A \Rightarrow B$ is true and $B \Rightarrow A$ is false, then A and B are not equivalent.

(3) If A is false, then $A \Rightarrow B$ is always true, for any B.

3.22 Suppose $P \Rightarrow Q$ and $Q \Rightarrow R$. Prove $P \Rightarrow R$. (This is called the **law of syllogism.)

*3.23 Let P, Q, R be propositions. Using truth tables or logical rules, show that

$$(P \Rightarrow Q) \wedge (P \Rightarrow R) \quad \Leftrightarrow \quad P \Rightarrow (Q \wedge R). \tag{3.3}$$